ELEC70013
EE9-AO4

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE PART IV: MEng and ACGI

HIGH PERFORMANCE ANALOGUE ELECTRONICS

Monday, 13 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	E. Rodriguez-Villegas
	Second Marker(s) :	S. Lucyszyn

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The Questions

(All of them are mandatory)

1. (a) A junior designer is given the task to create a Hartley receiver. Figure 1 shows the block diagram of the first design the designer creates. Find the output equations for the circuit in Figure 1.

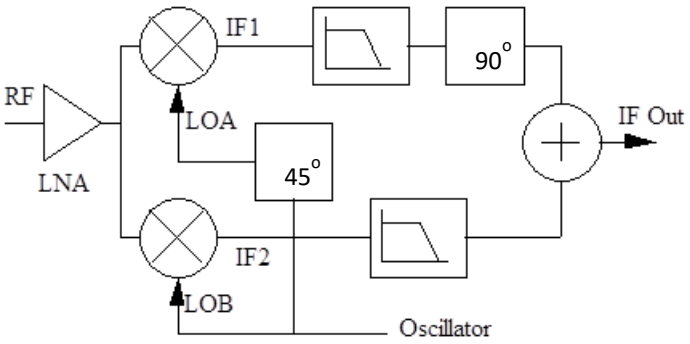


Figure 1

[8]

(b) Discuss what would be the ideal behaviour of the circuit in Figure 1, and whether this has been achieved by this design. You can assume all the blocks are ideal.

[4]

(c) Would you make any change to the design in Figure 1? Justify your answer.

[4]

(d) Briefly discuss what would be potential problems associated to not having an ideal oscillator.

[4]

2. (a) Explain briefly what is a disadvantage of having to add feedback to compensate for common mode variations in a transconductor.

[3]

(b) In a transconductor, explain what specification parameter will suffer if the transconductance is common mode dependent.

[3]

(c) Assuming that the transconductor in Figure 2 has been properly designed, would it suffer from the problem described in (b)? Briefly explain your answer.

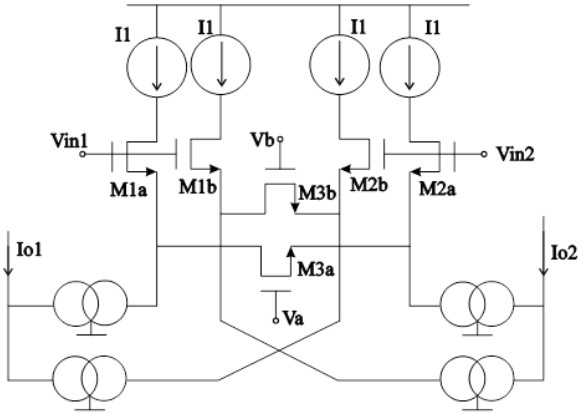


Figure 2

[3]

(d) A junior designer sends the circuit in Figure 2 for fabrication and when testing it, they observe a really high second order harmonic. A senior designer revises the design files and immediately a very obvious design mistake is found. What could that be?

[3]

(e) Even under ideal conditions, is the circuit in Figure 3 linear in large signal? Briefly explain your answer.

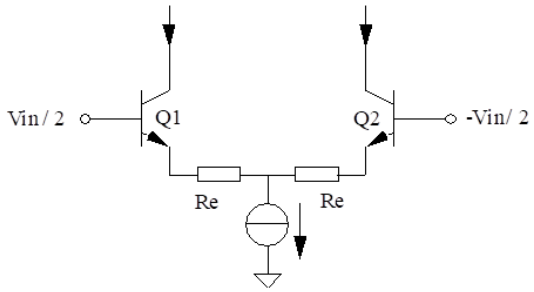


Figure 3

[4]

(f) Derive the approximate transconductance for the circuit in Figure 3. Explain your assumptions if any.

[4]

3. (a) Could there be any dominant device in terms of noise in the circuit in Figure 4?

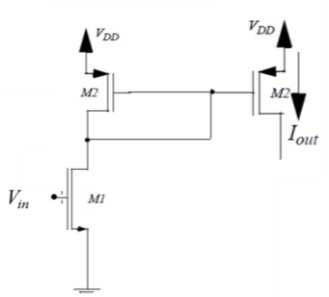


Figure 4

[4]

(b) Give an expression for the dominant output noise of the circuit in Figure 4, assuming very low frequency of operation (Hz range) and very large area transistors in the current mirror.

[4]

(c) What would be the dominant noise components for the circuit in Figure 4 if this had to operate at high speed and the aspect ratio (W/L) of the input transistor was very large compared to the ones in the current mirror. You can just give a descriptive answer with no equations.

[4]

(d) Give an example of a transconductor that could be used as a mixer, or alternatively of a combination of transconductors that could be used as mixers.

[4]

(e) What would be the dominant distortion harmonic in an ideal fully differential transconductor. Briefly explain your answer.

[4]

4. (a) Draw the schematic of an integrator built with the transconductor below, in Figure 5, (Hint: Don't forget to add biasing transistors).

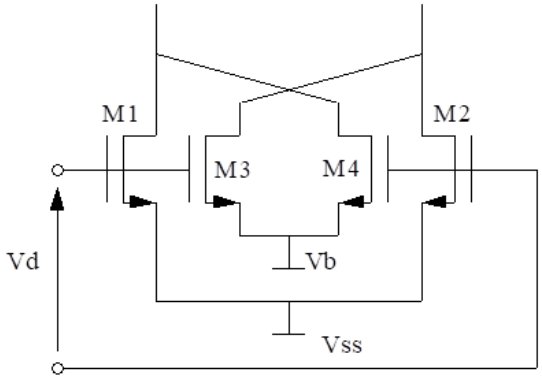


Figure 5

[4]

(b) What would be an advantage and a disadvantage of using transconductors versus amplifiers to create integrators for a filter?

[4]

(c) If a transconductor similar to the one above, in Figure 5, was to be designed but four terminal NMOS devices were not available, what would you do differently and why?

[3]

(d) Find an expression for the output current of the circuit proposed in (c) (Note: if you don't know the answer for (c) you can find the expression for the output current of the integrator drawn as the solution to (a) instead).

[3]

(e) How would you design the capacitors at the output of an integrator designed with the transconductor in (a) to minimise area whilst reducing mismatch in the differential branches at high frequencies?

[3]

(f) Explain how you would build a gyrator with the transconductor in Figure 5.

[3]